```
%PREPROCESSING

data = load('2.mat');
img = data.cjdata.image;
imshow(img,[]);
title('Original MRI Image');
img = mat2gray(img);
img_resized = imresize(img,[256 256]);
img_clahe = adapthisteq(img_resized);
img_filtered = medfilt2(img_clahe,[3 3]);
figure
subplot(1,2,1), imshow(img), title('Original MRI:')
subplot(1,2,2), imshow(img_filtered), title('Preprocessed Image:')

%SEGMENTATION

level = graythresh(img_filtered);
bw = imbinarize(img_filtered, level);
figure, imshow(bw), title('Otsu Binary');
bw = bwareaopen(bw, 200);
figure, imshow(bw), title('Noise Removed');
bw = imfill(bw,'holes');
figure, imshow(bw), title('Hole Filled');
se = strel('disk',15);
bw2 = imerode(bw,se);
figure, imshow(bw2), title('Skull Removed');
highInt = img_filtered > 0.55;   % tumor is very bright
figure, imshow(highInt), title('High Intensity Region');
highInt = highInt & bw2;
 % restrict inside brain
figure, imshow(highInt), title('Tumor Candidate');
highInt = bwareaopen(highInt,50);
se = strel('disk',3);
tumorMask = imclose(highInt,se);
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```
tumorMask = imfill(tumorMask,'holes');
se = strel('disk',4);
tumorMask = imdilate(tumorMask,se);
cc = bwconncomp(tumorMask);
numPixels = cellfun(@numel,cc.PixelIdxList);
[~,idx] = max(numPixels);   % tumor cluster is largest bright region
cleanMask = false(size(tumorMask));
cleanMask(cc.PixelIdxList{idx}) = true;
figure, imshow(cleanMask), title('Final Tumor Mask');
tumor = img_filtered .* cleanMask;
figure
subplot(1,2,1), imshow(img_filtered), title('Preprocessed')
subplot(1,2,2), imshow(tumor), title('Final Segmented Tumor')

%FEATURE EXTRACTION USING GLCM

tumor_gray = im2uint8(mat2gray(tumor));
tumor_gray(tumorMask == 0) = 0;
glcm = graycomatrix(tumor_gray, ...
    'Offset',[0 1; -1 1; -1 0; -1 -1], ...
    'Symmetric', true);
stats = graycoprops(glcm, ...
    {'Contrast','Correlation','Energy','Homogeneity'});
contrast     = mean(stats.Contrast);
correlation  = mean(stats.Correlation);
energy       = mean(stats.Energy);
homogeneity  = mean(stats.Homogeneity);
entropyValue = entropy(tumor_gray);
fprintf('\n---- GLCM Features ----\n');
fprintf('Contrast     : %.4f\n', contrast);
fprintf('Correlation  : %.4f\n', correlation);
fprintf('Energy       : %.4f\n', energy);
fprintf('Homogeneity  : %.4f\n', homogeneity);
fprintf('Entropy      : %.4f\n', entropyValue);
```